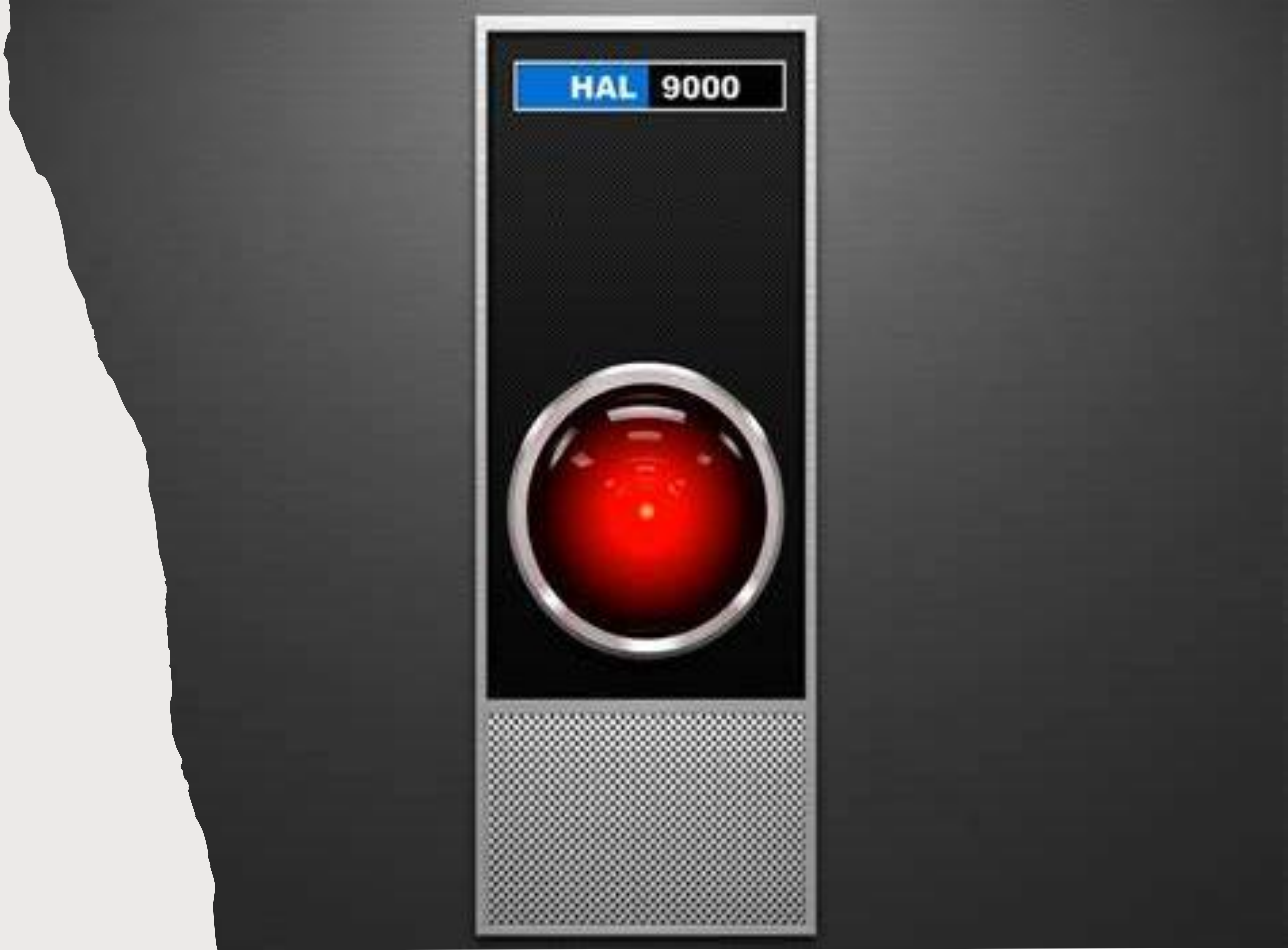
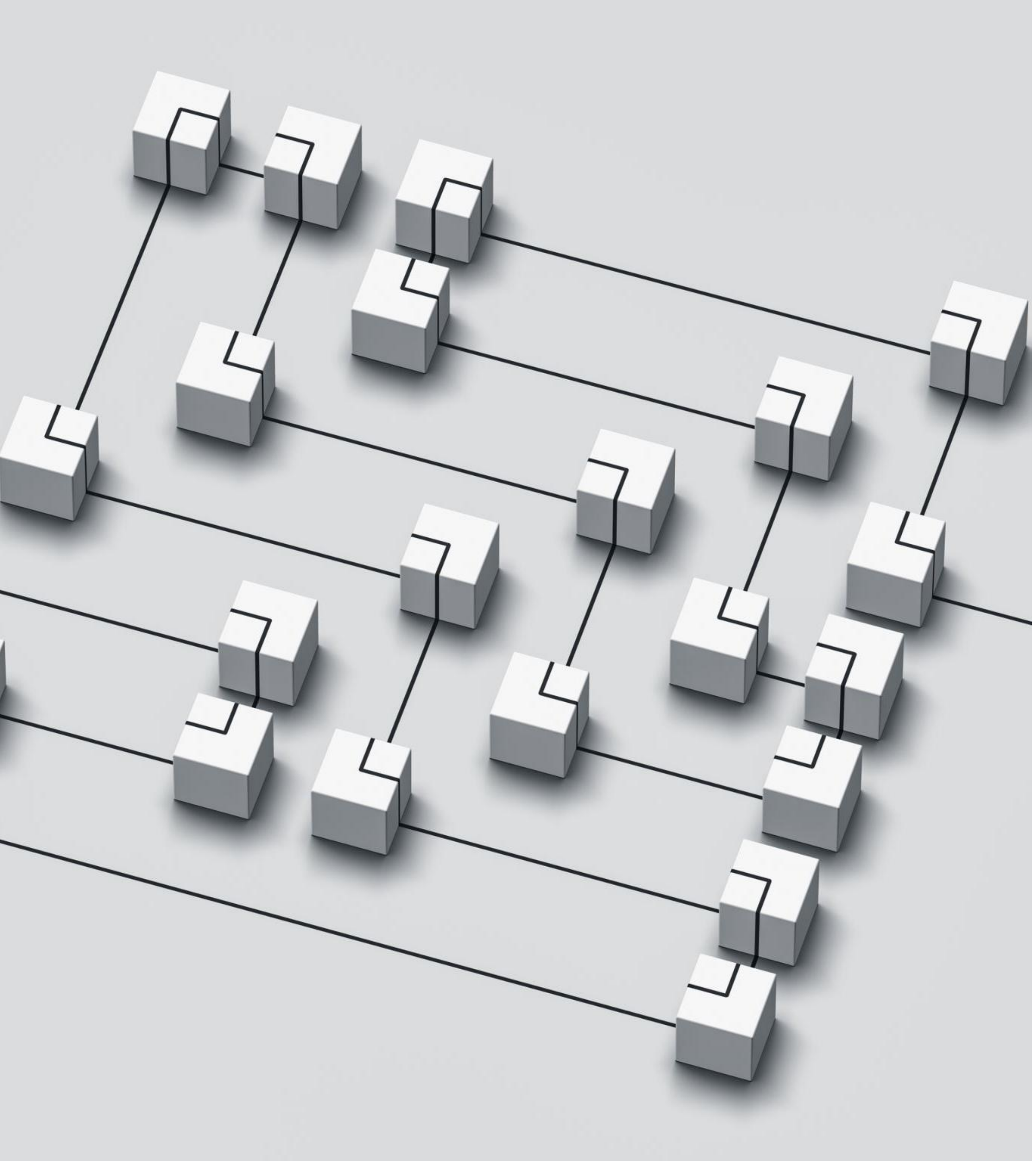
Understanding AI and Prompting Techniques

What is AI?





AI Terms

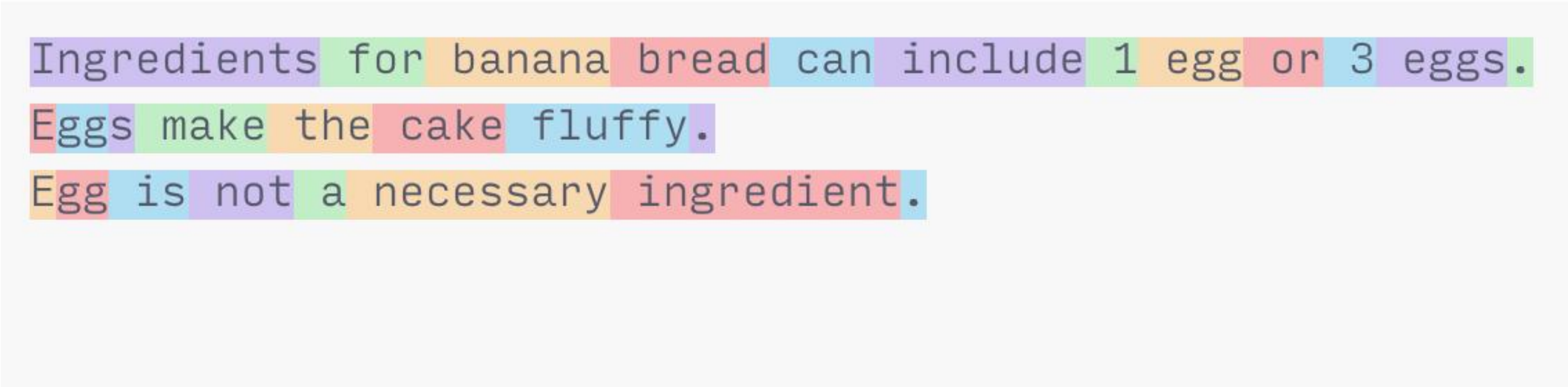
- Models
- Tokens
- Prompts
- Context
- Roles
- Agents
- Plugins

Models

- An **AI model** is a system designed to process data and make predictions, decisions, or generate content based on learned patterns. AI models are trained using large datasets and advanced algorithms to recognize trends, solve problems, and automate tasks.

COMPANY	MODEL
Microsoft	GPT-4o, DALL-E, Phi-4
Meta	LLaMA 2, LLaMA 3
Anthropic	Claude Sonnet 3.7
DeepSeek	DeepSeek R1

Tokens



In AI and natural language processing, tokens are the individual units of text that a model processes. These tokens can be words, parts of words, or even characters, depending on how the AI model is designed.

[Tokenizer - OpenAI API]

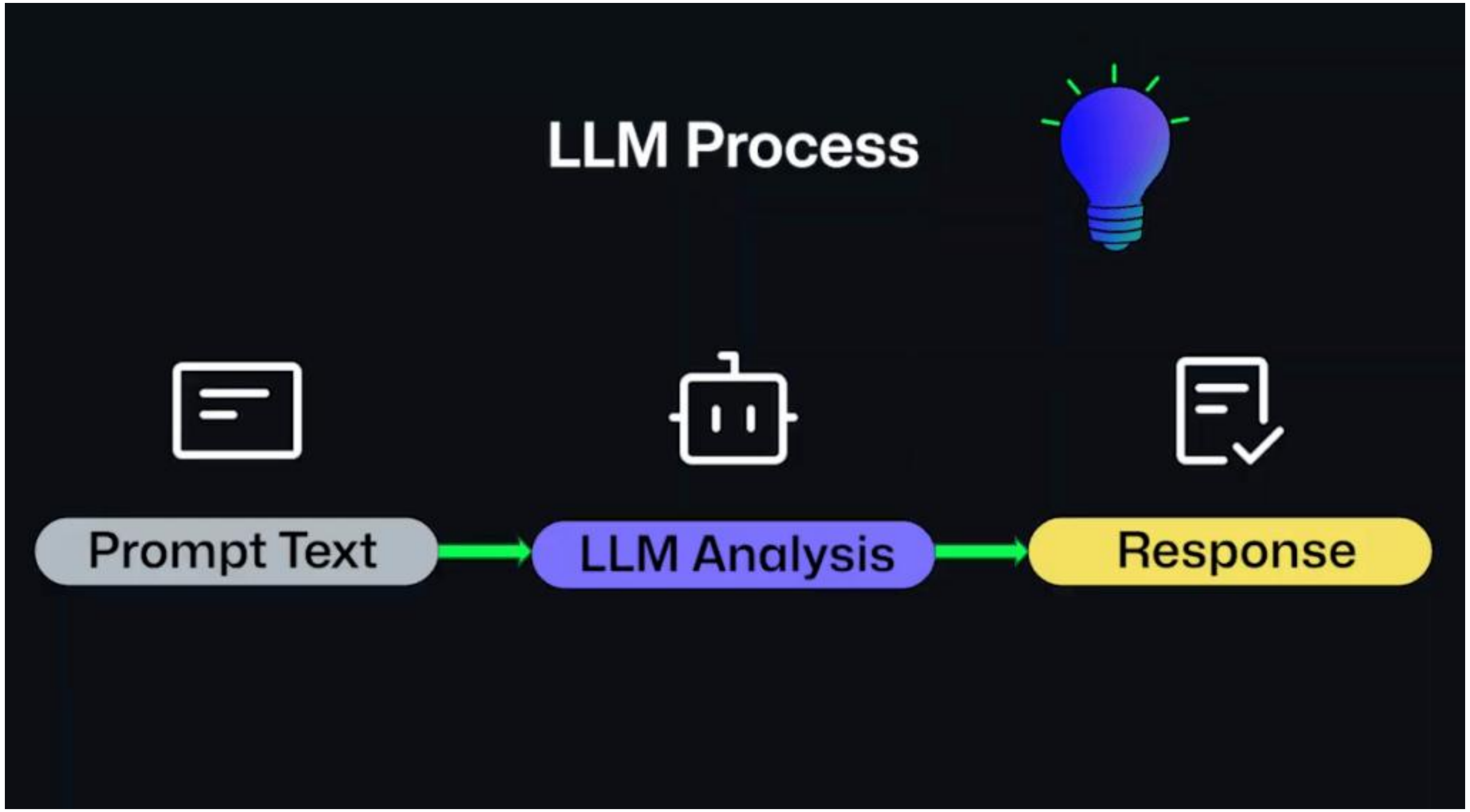
1 token ~= 4 chars in English
1 token ~= ¾ words
100 tokens ~= 75 words

Prompt Engineering

Prompt Engineering is about effective communication

Exact Instructions

LLMs



Prompts

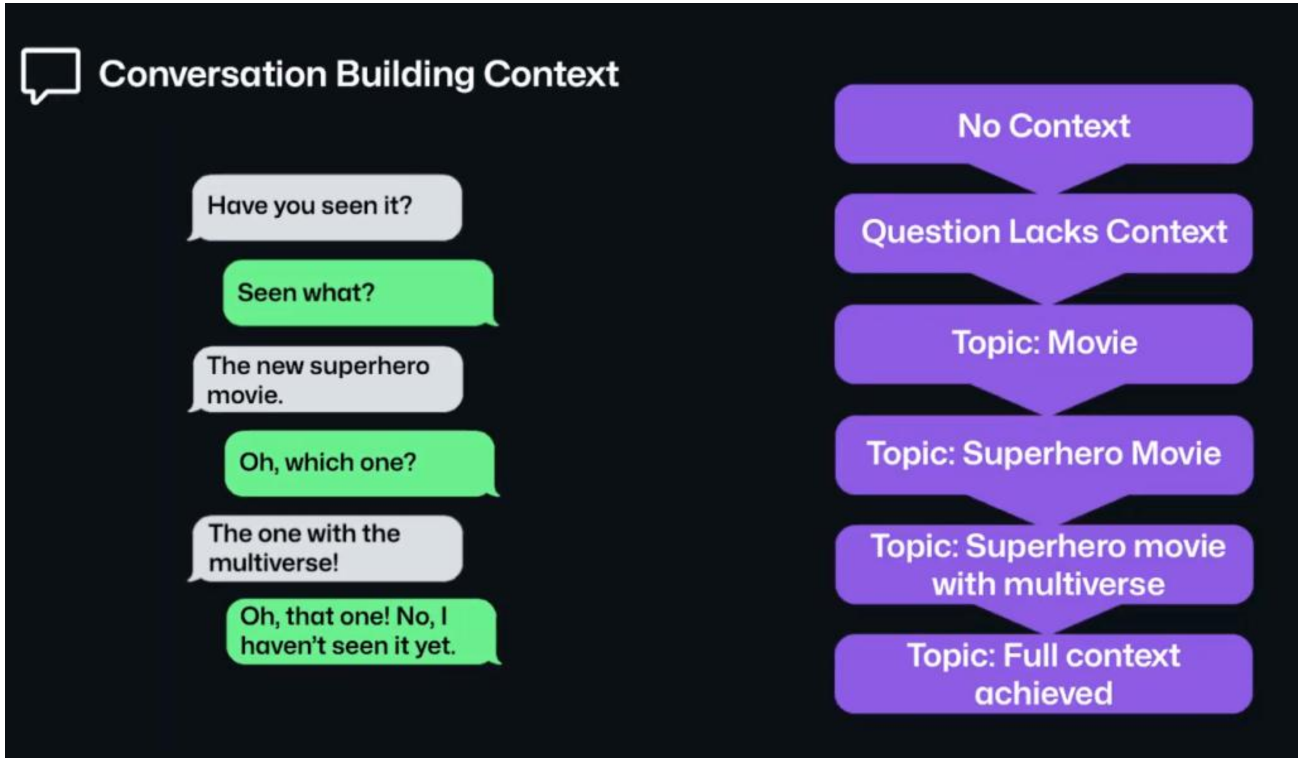
System Prompt

- A predefined instruction that shapes the AI’s behavior or tone.
- Example: "Always respond in a professional and concise manner."
- These are used to guide how the AI interacts with users.

User Prompt

- The direct input a user provides to an AI system.
- Example: "Summarize the benefits of AI in education."
- The AI responds based on the user's request.

Context


Conversation Building Context
Have you seen it?
Seen what?
The new superhero movie.
Oh, which one?
The one with the multiverse!
Oh, that one! No, I haven't seen it yet.
No Context
Question Lacks Context
Topic: Movie
Topic: Superhero Movie
Topic: Superhero movie with multiverse
Topic: Full context achieved

Context in AI is crucial because it helps the system generate more accurate and relevant responses.

Prompt without context:

"Write an email inviting people to a meeting."

Subject: **Meeting Invitation**

Hi everyone,

I'd like to invite you all to a meeting. Please confirm your availability.

- Best regards,
  [Your Name]

Context in AI is crucial because it helps the system generate more accurate and relevant responses.

Prompt without context:

"I'm leading the Azure Resiliency project at Microsoft. Write an email inviting my Azure Reliability team to a kickoff meeting scheduled for next Thursday at 2 PM ET. The meeting will discuss goals for improving our outage analysis using AI and the Semantic Kernel."

Subject: **Kickoff Meeting: Azure Resiliency Project – AI & Semantic Kernel**

Hi Azure Reliability Team,

I'm excited to invite you to our project kickoff meeting scheduled for next **Thursday at 2 PM ET**. We'll discuss our objectives for enhancing outage analysis by integrating advanced AI techniques and leveraging Microsoft's Semantic Kernel.

Agenda:

- Overview of project goals
- Introduction to AI-driven outage analysis strategies
- Role of Semantic Kernel in our methodology
- Next steps and action items

Please confirm your attendance, and feel free to reach out with any questions or suggestions.

Looking forward to our collaboration!

- Best regards,
  [Your Name]
  Azure Resiliency Lead

Prompting techniques

Not throwing away my..shot.

Zero-Shot Prompting

Definition: You give the model only an instruction, with no examples.

When to use: Quick tasks where the model's pre-training covers enough knowledge to perform accurately.

Example:

- Prompt: "Summarize the plot of The Little Prince in two sentences."
- What happens: The model relies entirely on its internal knowledge to generate the summary.

One-Shot Prompting

Definition: You provide the instruction plus one example demonstration.

When to use: When you want the model to see your desired format or style before generating its own response.

Prompt: "Convert the following casual message into a formal email style.
Example:
'Hey team, I'm running late to the meeting—caught in traffic!' →
'Dear all, I regret to inform you that I will arrive late to today's meeting due to unexpected traffic delays.'

Now convert:
'Just got your text—sounds good. Let's catch up later.' →"

Few-Shot Prompting

Definition: You supply the instruction along with several (typically 3–10) example input–output pairs.

When to use: For more complex tasks or when you need the model to learn a pattern or style variation that one example can't fully convey.

Prompt:
"Rewrite each casual date into ISO format (YYYY-MM-DD).
Examples:
• 'We'll meet next Friday.' → '2025-05-09'
• 'Her birthday is on July 4th.' → '2025-07-04'
• 'The report is due the first Monday of June.' → '2025-06-02'
Now convert:
• 'I started the project two weeks ago.' →"

Roles

- **Defines Persona & Expertise**
  - By saying “You are a security specialist” or “Act as a senior Python tutor,” you instantly give the model a knowledge domain and voice to adopt.
- **Sets Tone & Style**
  - A “friendly travel guide” will write differently than a “formal legal advisor.” Roles lock in register (formal vs. casual), pacing, and vocabulary.
- **Reduces Ambiguity**
  - Clear expectations (“You are a requirements engineer, ask clarifying questions first”) guide the model to take the right next step rather than guessing.
- **Establishes Guardrails**
  - Roles can embed soft constraints (“You are a safety-critical systems reviewer—flag any potential issues”) that help prevent irrelevant or unsafe outputs.

Roles

Cloud Solutions Architect	“You are a Cloud Solutions Architect specializing in Azure. Review this ARM template and recommend improvements for scalability and cost optimization.”
DevOps Pipeline Engineer	“You are a DevOps engineer expert in GitHub Actions. Critique this workflow YAML for building, testing, and deploying a .NET Core app.”
Kubernetes SRE	“You are a Site Reliability Engineer focused on Kubernetes. Analyze these pod logs and suggest fixes for the recurring OOMKilled errors.”
Security Compliance Auditor	“You are a cloud security auditor. Evaluate this Bicep deployment for adherence to CIS benchmark controls.”
Performance Tuning Specialist	“You are a .NET performance expert. Profile this ASP.NET app under load and advise on hotspots and GC tuning.”
API Documentation Writer	“You are an API technical writer. Generate OpenAPI-style documentation for this new set of REST endpoints.”
Backend Microservices Debugger	“You are a microservices architect. Diagnose the latency spike between Service A and Service B and recommend tracing strategies.”

Roles

Higher Relevance & Accuracy

Models tuned to a role draw on the right knowledge and formatting conventions, cutting down on off-topic drifts.

Consistent Voice Across Interactions

Especially in long chats or multi-turn workflows, a defined role keeps tone and style uniform.

Faster Prompt Iteration

Once your role prompt is solid, you only tweak the specifics of each request—not the persona setup every time.

Improved Safety & Compliance

Roles can embed ethical or regulatory reminders (e.g., “You are a GDPR-aware data processor—do not reveal personal data.”).

Scalability in Complex Pipelines

In agentic systems, you can spin up specialized sub-agents (e.g., a “retriever agent” vs. a “reasoning agent”) each with its own role, making architectures modular and easier to manage.

Prompt Patterns

RTF	TAG	BAB	CARE	RISE
• Act as a [ROLE] • Create a [TASK] • Show as [FORMAT]	• Define [TASK] • State the [ACTION] • Clarify the [GOAL]	• Explain the Problem [BEFORE] • State the Outcome [AFTER] • Ask for the [BRIDGE]	• Give the [CONTEXT] • Describe the [ACTION] • Clarify the [RESULT] • Give the [EXAMPLE]	• Specify the [ROLE] • Describe the [INPUT] • Ask for [STEPS] • Describe the [EXPECTATION]

Prompt Best Practices

An effective prompt is **clear and precise**, because ambiguity can confuse the model.

It's also important to provide *enough* **context**, but not too much detail, since this can overwhelm the LLM.

If you don't get the answer you're expecting, don't forget to **iterate and refine** your prompts!

Challenges in AI Prompt Engineering

- Variability (Non Deterministic)
  - Same Prompt, Different Outputs
- Fabrication (aka Hallucination)
  - AI Can Invent Plausible but False Information
- Prompt confusion
  - Mixing multiple requests or being unclear when writing prompts
- Assumption Errors
  - Assuming the LLM has context it doesn’t have
- Limitations
  - Trying to give too much information or ask for too much

Patterns

Retrieval-Augmented Generation (RAG)	Tool-Use Pattern	Autonomous Agents
• Combines an LLM with a vector-based document store – AI Search	• Wraps an LLM in a "planner + executor" loop	• Multi-step bots that self-drive toward a goal

Prompting Techniques | Prompt Engineering Guide

Prompting Techniques

Prompt Engineering helps to effectively design and improve prompts to get better results on different tasks with LLMs.

While the previous basic examples were fun, in this section we cover more advanced prompting engineering techniques that allow us to achieve more complex tasks and improve reliability and performance of LLMs.

Zero-shot Prompting	Few-shot Prompting	Chain-of-Thought Prompting
Meta Prompting	Self-Consistency	Generate Knowledge Prompting
Prompt Chaining	Tree of Thoughts	Retrieval Augmented Generation
Automatic Reasoning and Tool-use	Automatic Prompt Engineer	Active-Prompt
Directional Stimulus Prompting	Program-Aided Language Models	ReAct
Reflexion	Multimodal CoT	Graph Prompting

Copilot

Microsoft 365
Copilot

Copilot for Work

Has context about:

Your calendar

Your email

Your work files

Meetings

Support for Agents

Researcher

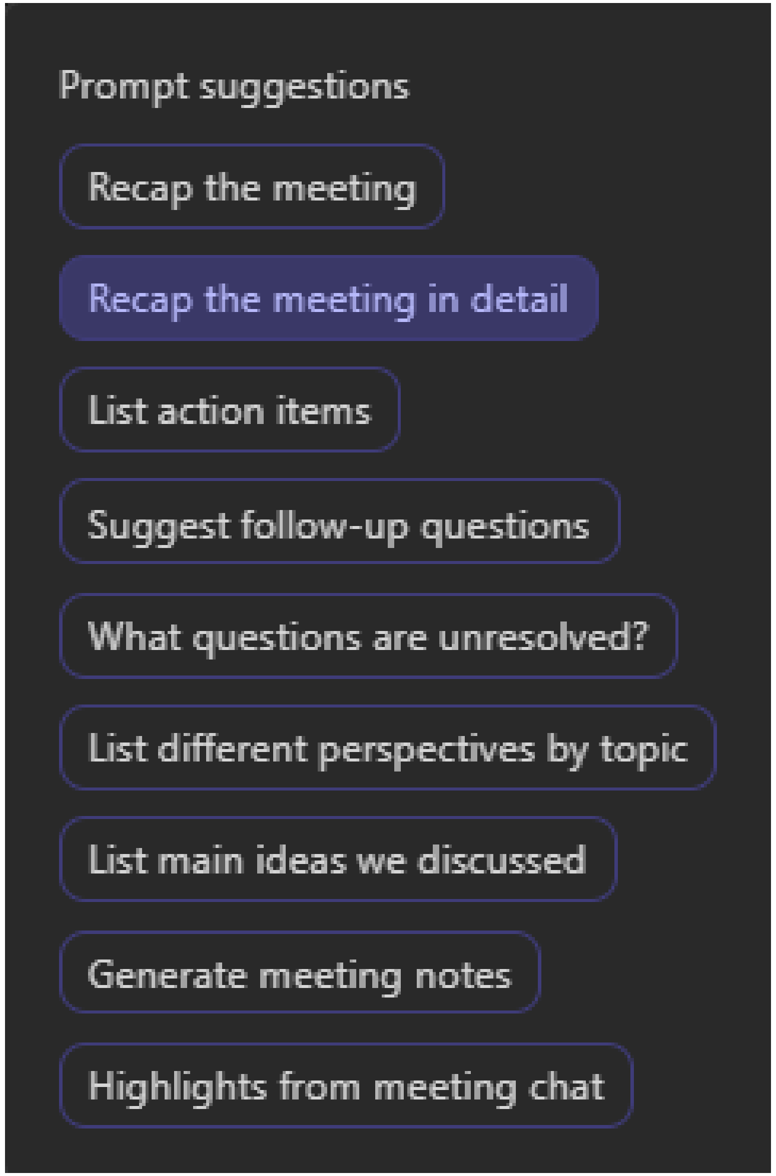
Analyst

Writing Coach

Other custom agents (KAI, Engineering Wiki)

Copilot in Teams

Transcription is required to use Copilot.



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